Year 9 Mathematics

Unit 2: Area and Volume

2.6 Pythagoras' Theorem

Pythagoras' Theorem

STARTER · 5 MIN

Date: _____ **Title: Pythagoras' Theorem**

LEARNING OBJECTIVES

- Find the hypotenuse of a right-angled triangle
- Find a shorter side of a right-angled triangle
- Check for a right angle and find the distance between two points

STARTER *Answer in your book. No calculators.*

1. Work out 9^2

2. Work out $\sqrt{144}$

3. Work out $3^2 + 4^2$

4. Work out $13^2 - 5^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. **81**

2. **12**

3. **25**

4. **144**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the hypotenuse of a right-angled triangle	☐
SILVER	I can find a shorter side of a right-angled triangle	☐
GOLD	I can check for a right angle and find the distance between two points	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Pythagoras' Theorem

INTRODUCTION · 3 MIN

In a right-angled triangle, the hypotenuse is the longest side, opposite the right angle.

The theorem

The square on the hypotenuse c equals the sum of the other two.

$a^2 + b^2 = c^2$

Hypotenuse

Add the squares, then square root.

$c^2 = 6^2 + 8^2 = 100,\ \ c = 10$

Shorter side

Subtract the squares, then square root.

$a^2 = 13^2 - 5^2 = 144,\ \ a = 12$

Right angle?

Check if the two shorter sides fit.

$5^2 + 12^2 = 13^2$, so yes

Examples 1 and 2 — Finding the Hypotenuse

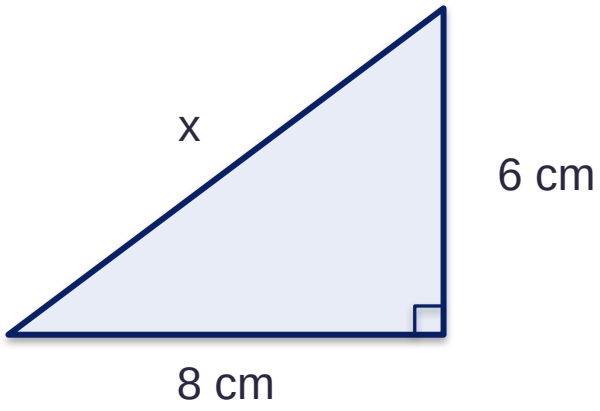
EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint $c^2 = a^2 + b^2$ c is the hypotenuse

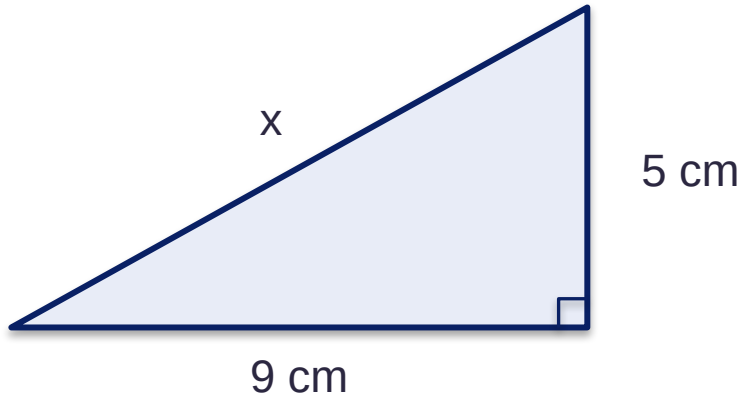
Example 1

Find x.



Example 2

Find x. Give your answer to 3 s.f.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Hypotenuse: add the squares

$x^2 = 8^2 + 6^2 = 64 + 36 = 100$

$x = \sqrt{100}$

x = 10 cm

Tip: x is opposite the right angle.

Example 2

Hypotenuse: add the squares

$x^2 = 9^2 + 5^2 = 81 + 25 = 106$

$x = \sqrt{106} = 10.295\ldots$

x = 10.3 cm (3 s.f.)

Tip: the hypotenuse is the longest side.

Task 1

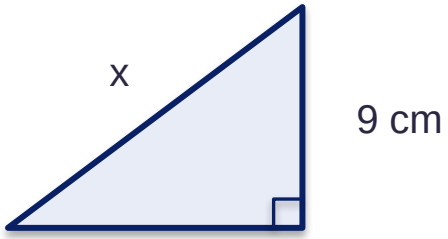
PRACTICE · 9 MIN

BRONZE

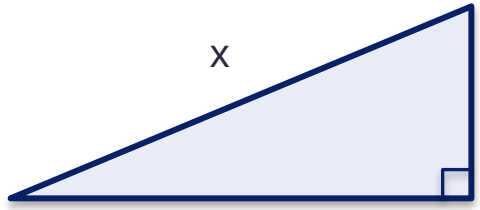
Find x. Give your answers to 3 s.f. where needed.

Hint $c^2 = a^2 + b^2$ c is the hypotenuse

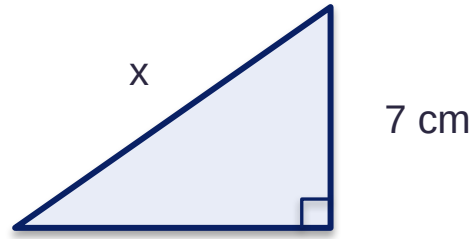
1 Find x



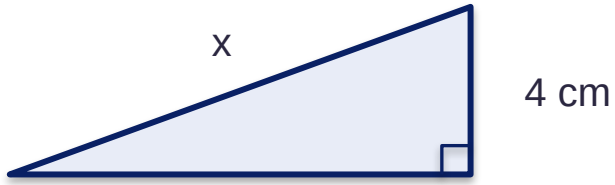
2 Find x



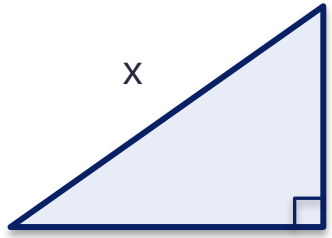
3 Find x



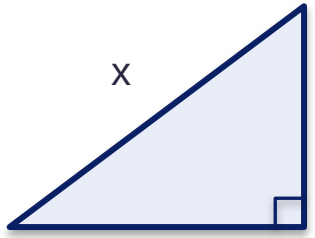
4 Find x



5 Find x



★ Find x



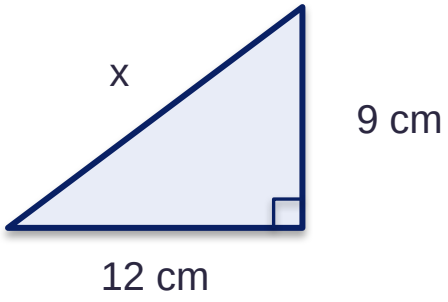
Task 1 — Answers

CORRECTIONS

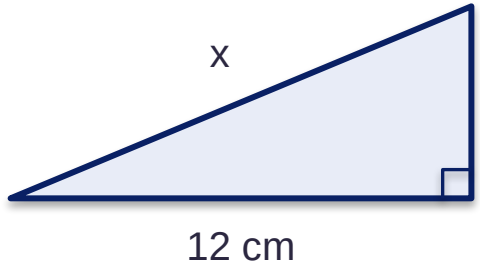
BRONZE

Tick or correct in green pen

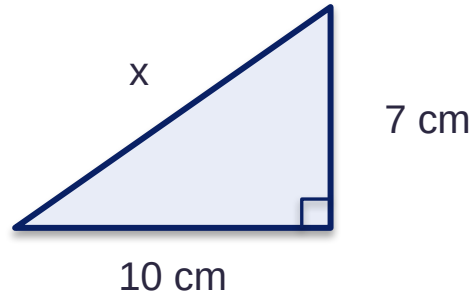
1 **x = 15 cm**



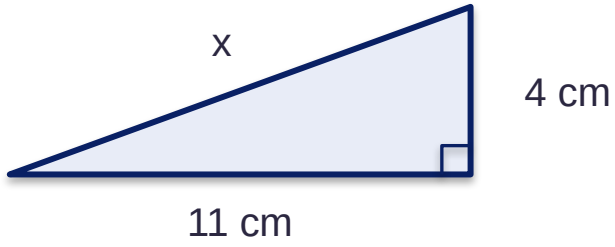
2 **x = 13 cm**



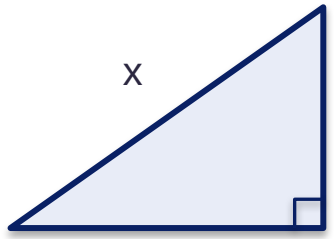
3 **x = 12.2 cm**



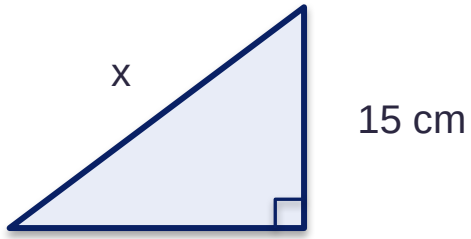
4 **x = 11.7 cm**



5 **x = 10.4 cm**



★ **x = 25 cm**



Examples 3 and 4 — Finding a Shorter Side

EXAMPLES · 5 MIN

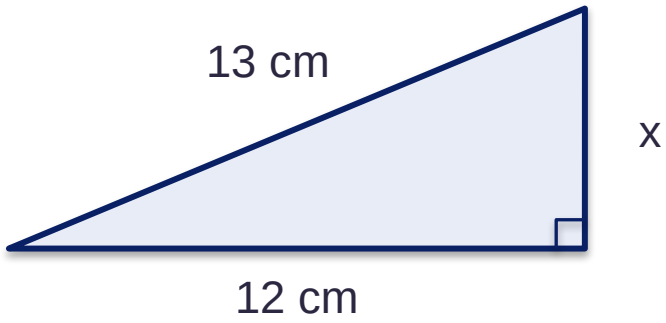
SILVER

Watch and copy into your book

Hint $a^2 = c^2 - b^2$ c is the hypotenuse

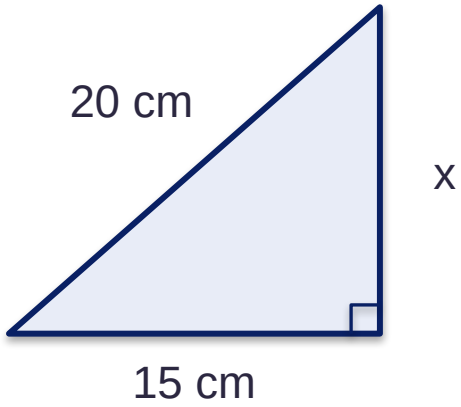
Example 3

Find x.



Example 4

Find x. Give your answer to 3 s.f.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Shorter side: subtract

$x^2 = 13^2 - 12^2 = 169 - 144 = 25$

$x = \sqrt{25}$

x = 5 cm

Tip: hypotenuse² − other side²

Example 4

Shorter side: subtract

$x^2 = 400 - 225 = 175$

$x = \sqrt{175} = 13.228\ldots$

x = 13.2 cm (3 s.f.)

Tip: x must be shorter than 20 cm.

Task 2

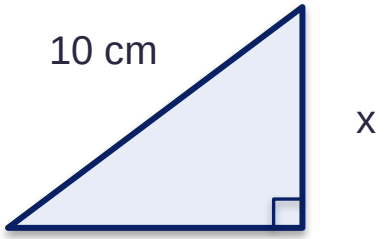
PRACTICE · 9 MIN

SILVER

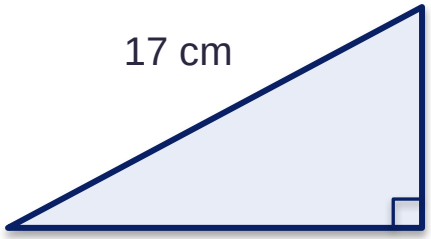
Find x. Give your answers to 3 s.f. where needed.

Hint $a^2 = c^2 - b^2$ c is the hypotenuse

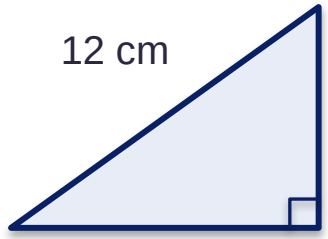
1 Find x



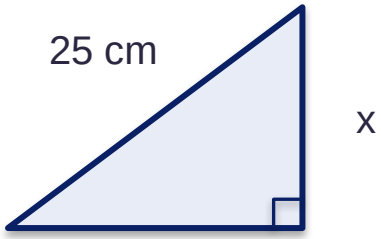
2 Find x



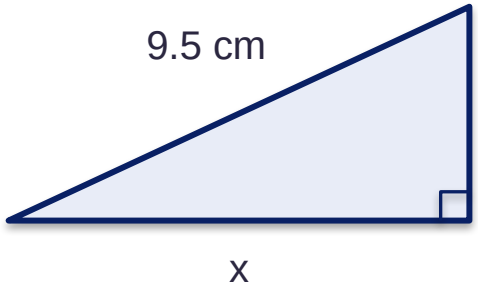
3 Find x



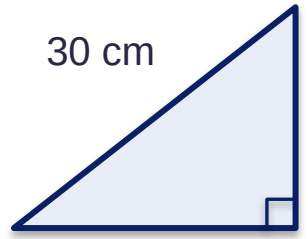
4 Find x



5 Find x



★ Find x



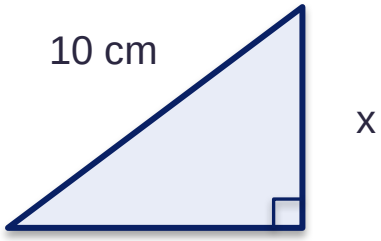
Task 2 — Answers

CORRECTIONS

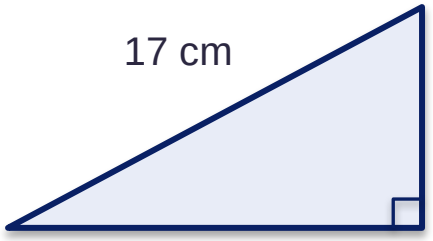
SILVER

Tick or correct in green pen

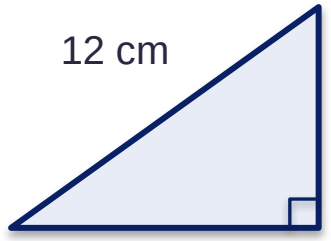
1 **x = 6 cm**



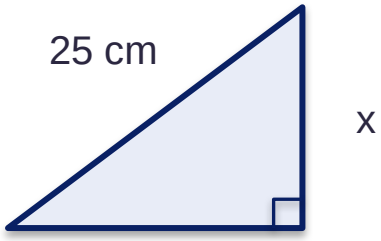
2 **x = 8 cm**



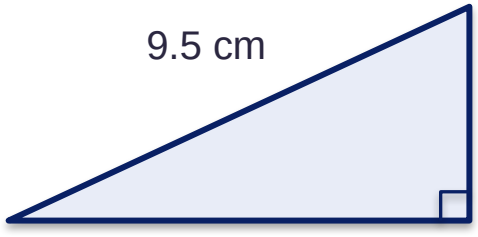
3 **x = 9.75 cm**



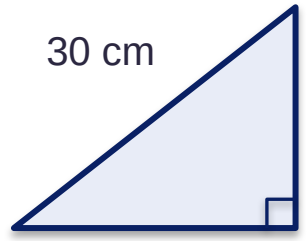
4 **x = 15 cm**



5 **x = 8.62 cm**



★ **x = 23.6 cm**



Examples 5 and 6 — Right Angles and Distances

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint right-angled if $a^2 + b^2 = c^2$ distance: across^2 + up^2

Example 5

(2 marks)

A triangle has sides 8 cm, 15 cm and 17 cm.

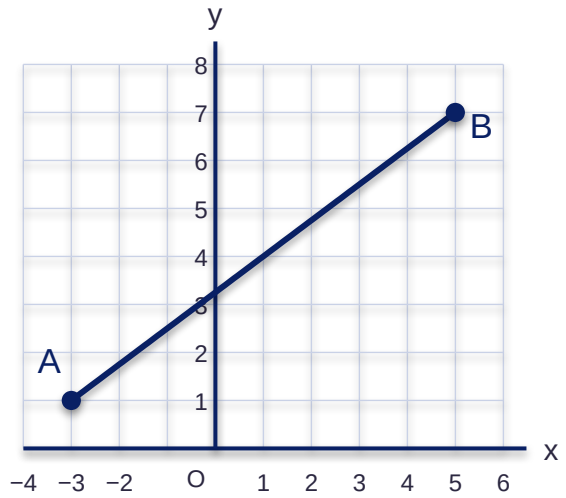
Is it right-angled?

Give a reason.

Example 6

(3 marks)

Find the distance between A(−3, 1) and B(5, 7).



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Do the shorter sides fit?

$8^2 + 15^2 = 64 + 225 = 289$

$17^2 = 289$

Yes: $8^2 + 15^2 = 17^2$

Tip: the longest side would be c.

Example 6

Across and up, then Pythagoras

across: $5 - (-3) = 8$

up: $7 - 1 = 6$

$AB^2 = 8^2 + 6^2 = 100$

AB = 10 units

Tip: sketch the right-angled triangle.

Task 3

PRACTICE · 9 MIN

GOLD

Questions 1–3: is it right-angled? 4–6: distance between the points

Hint right-angled if $a^2 + b^2 = c^2$
distance: across^2 + up^2

1. Sides 5 cm, 12 cm, 13 cm

2. Sides 6 cm, 7 cm, 9 cm

3. Sides 10 cm, 10 cm, 14 cm

4. (2, 3) and (5, 7)

5. (−4, −1) and (8, 4)

★ (−6, 2) and (3, −3) (3 s.f.)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. Yes: 25 + 144 = 169
2. No: 36 + 49 = 85, not 81
3. No: 100 + 100 = 200, not 196
4. 5 units
5. 13 units

★ $\sqrt{106}$ = 10.3 units

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the hypotenuse: shorter sides 3 cm and 4 cm

SILVER Hypotenuse 15 cm, one side 9 cm. Find the other side.

GOLD Find the distance between (1, 1) and (4, 5).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **5 cm**

SILVER **12 cm**

GOLD **5 units**

Exam Practice — The Hypotenuse

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

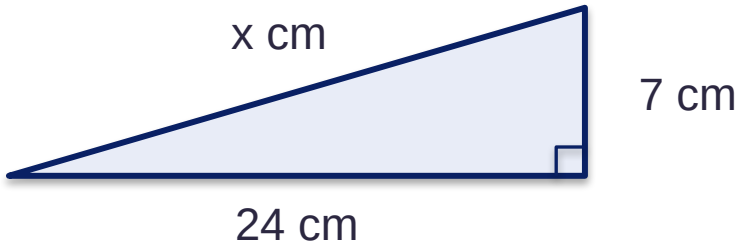
Hint $c^2 = a^2 + b^2$ c is the hypotenuse

Question 1

(2 marks)

Find the length of the side marked x.

Diagram NOT accurately drawn



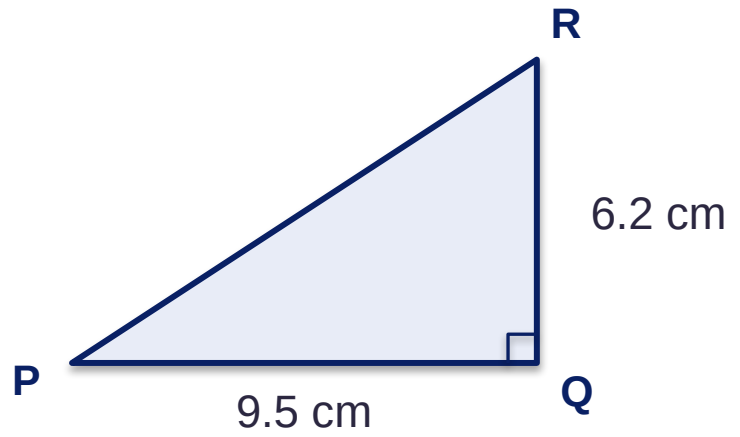
..... cm

Question 2

(3 marks)

Find the length of PR.
Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... cm

Exam Practice — A Shorter Side

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

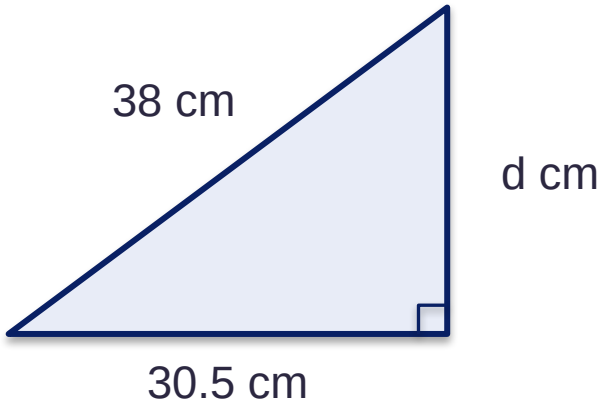
Hint $a^2 = c^2 - b^2$ c is the hypotenuse

Question 3

(3 marks)

Find the length of the side marked d. Give your answer to 3 s.f.

Diagram NOT accurately drawn



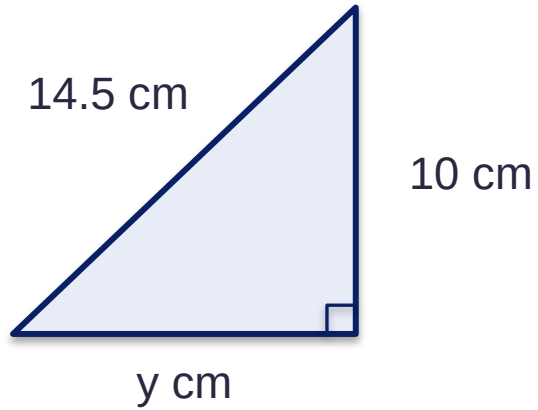
..... cm

Question 4

(2 marks)

Find the length of the side marked y.

Diagram NOT accurately drawn



..... cm

Exam Practice — Right Angles and Distances

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

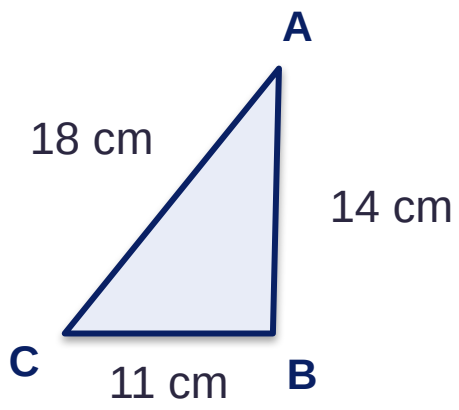
Hint right-angled if $a^2 + b^2 = c^2$
distance: across^2 + up^2

Question 5

(2 marks)

Ahmed says, 'The triangle ABC is a right-angled triangle.'
Is he correct? Give a reason.

Diagram NOT accurately drawn



Question 6

(3 marks)

Find the distance between the points P(−10, −3) and Q(2, 2).
You could use a sketch to help.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$x^2 = 24^2 + 7^2 = 625$ — M1

x = 25 cm — A1

Tip: add the squares for the hypotenuse.

Question 2

$6.2^2 + 9.5^2 = 128.69$ — M1

$\sqrt{128.69} = 11.344\ldots$ — M1

PR = 11.3 cm — A1

Tip: PR is opposite the right angle.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$38^2 - 30.5^2 = 513.75$ — M1

$\sqrt{513.75} = 22.66\ldots$ — M1

d = 22.7 cm — A1

Tip: subtract to find a shorter side.

Question 4

$14.5^2 - 10^2 = 110.25$ — M1

y = 10.5 cm — A1

Tip: check y is shorter than 14.5 cm.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$11^2 + 14^2 = 121 + 196 = 317$ — B1

$18^2 = 324$, so no: $317 \neq 324$ — B1

Tip: test the two shorter sides.

Question 6

across 12, up 5 — M1

$12^2 + 5^2 = 169$ — M1

PQ = 13 — A1

Tip: $2 - (-10) = 12$ and $2 - (-3) = 5$